

JEE MAIN REVISION MASTER: ELECTROCHEMISTRY

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 35 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
High	Nernst Equation & Cell Potential	11 PYQs	Core Formula
High	Electrolytic Conductance & Cell Constant	9 PYQs	Conductance
High	Faraday's Laws of Electrolysis	9 PYQs	Quantitative Electrolysis
Moderate	Electrode Potential & Delta G Relation	4 PYQs	Thermodynamics
Standard	Kohlrausch's Law & Weak Electrolyte Dissociation	2 PYQs	Electrolyte Behavior
Standard	Commercial Batteries, Fuel Cells & Corrosion	1 PYQs	Applied Electrochemistry

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Nernst Equation & Cell Potential	$E_{\text{cell}} = E^{\circ}_{\text{cell}} - \frac{0.0591}{n} \log_{10} Q$, $\Delta G^{\circ} = -n F E^{\circ}_{\text{cell}} = -2.303 RT \log_{10} K$
Electrolytic Conductance & Cell Constant	$\kappa = \frac{1}{R} \left(\frac{l}{A} \right) = G \cdot G^*$, $\Lambda_m = \frac{1000 \cdot \kappa}{M}$ (κ in S cm^{-1})
Faraday's Laws of Electrolysis	$w = \frac{M \cdot I \cdot t}{n \cdot F}$, $\text{Moles of metal} = \frac{I \cdot t}{n \cdot F}$, $F \approx 96500 \text{ C mol}^{-1}$
Electrode Potential & Delta G Relation	$\Delta G = -nFE$, $\Delta G^{\circ}_3 = \Delta G^{\circ}_1 + \Delta G^{\circ}_2$ $\implies n_3 E^{\circ}_3 = n_1 E^{\circ}_1 + n_2 E^{\circ}_2$
Kohlrausch's Law & Weak Electrolyte Dissociation	$\Lambda_m^{\circ} = \nu_+ \lambda_+^{\circ} + \nu_- \lambda_-^{\circ}$, $\alpha = \frac{\Lambda_m}{\Lambda_m^{\circ}}$, $K_a = \frac{c \alpha^2}{1 - \alpha}$
Commercial Batteries, Fuel Cells & Corrosion	Lead Storage Discharge: $\text{Pb(s)} + \text{PbO}_2\text{(s)} + 2\text{H}_2\text{SO}_4 \rightarrow 2\text{PbSO}_4\text{(s)} + 2\text{H}_2\text{O}$, $\eta = \frac{\Delta G}{\Delta H}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Nernst Equation & Cell Potential (High Yield — 11 PYQs)

Core Principle: Calculation of non-standard cell potential $E_{\text{cell}} = E^{\circ}_{\text{cell}} - (0.0591 / n) \log_{10} Q$ at 298 K, equilibrium constant K_{eq} , and concentration cells where $E^{\circ}_{\text{cell}} = 0$.

■ **Caution / JEE Pitfall:** 1. Forgetting to raise concentrations to their stoichiometric powers in reaction quotient Q . 2. Confusing oxidation vs reduction potential signs: $E^{\circ}_{\text{(red)}} = -E^{\circ}_{\text{(ox)}}$. 3. In concentration cells, anode is always the dilute compartment ($c_1 < c_2$).

■ **Speed Trick:** At 298 K, $E^{\circ} = (0.0591 / n) \log_{10} K$. For spontaneous cell operation, $E_{\text{cell}} > 0$ and $\Delta G < 0$.

Representative Source PYQs:

- [2026 Shift] Q#2 (MCQ): "Given below are the half-cell reactions: $\text{Mn}^{2+} + 2e^- \rightarrow \text{Mn}$ ($E^{\circ} = -1.18\text{V}$) $2\text{Mn}^{3+} + e^- \rightarrow \text{Mn}^{2+}$ ($E^{\circ} = 1.50\text{V}$)" [Ans: N/A]
- [2025 Shift] Q#1 (MCQ): "The correct statement/s about Galvanic cell is/are (a) Current flows from cathode to anode (b) Anode is positive terminal (c) If $E_{\text{cell}} > 0$ " [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "For a given half cell, $\text{Al}^{3+} + 3e^- \rightarrow \text{Al}$ on increasing of aluminium ion, the electrode potential will..." [Ans: N/A]

3.2 Electrolytic Conductance & Cell Constant (High Yield — 9 PYQs)

Core Principle: Specific conductivity $\kappa = (1 / R) \left(l / A \right) = G \cdot G^*$. Molar conductivity $\Lambda_m = (1000 \kappa / M)$ (with κ in S cm^{-1}) or $\Lambda_m = (\kappa / 1000M)$ (with κ in S m^{-1}).

■ **Caution / JEE Pitfall:** Unit confusion between cm and m. If κ is in S m^{-1} , use $\Lambda_m = (\kappa / c)$ (c in mol m^{-3}). If in S cm^{-1} , use $\Lambda_m = (1000 \kappa / M)$ (M in mol L^{-1}).

■ **Speed Trick:** Cell constant $G^* = (l / A)$ is an intrinsic geometric property of the conductivity cell and is strictly independent of the electrolyte filled.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The conductivity of centimolar solution of KCl at 298 K is $0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing ..." [Ans: C]
- [2025 Shift] Q#2 (MCQ): "The electronic conductance depends on..." [Ans: N/A]
- [2023 Shift] Q#4 (MCQ): "The resistance of 0.1M weak acid HA in a conductivity cell is $2 \times 10^3 \text{ Ohm}$. The cell constant of the cell is 0.78 cm^{-1} and ..." [Ans: N/A]

3.3 Faraday's Laws of Electrolysis (High Yield — 9 PYQs)

Core Principle: First and second laws of quantitative electrolysis: Mass deposited $w = Z I t = (M \cdot I \cdot t / n F)$. Total charge passed $Q = I t = n_e F$.

■ **Caution / JEE Pitfall:** 1. Using an incorrect n -factor (electrons transferred per molecule). E.g. for O_2 from H_2O , $n = 4$; for H_2 , $n = 2$. 2. Time t must strictly be substituted in seconds (s).

■ **Speed Trick:** Gas volume at STP: Moles $\times 22.4 \text{ L}$. For electroplating, Volume = Area $\times$ Thickness and Mass = Density $\times$ Volume.

Representative Source PYQs:

- [2026 Shift] Q#4 (MCQ): "During the electrolysis of acidified water, 16 g of O_2 gas is formed at anode. The volume of H_2 gas liberated at c..." [Ans: A]
- [2025 Shift] Q#4 (MCQ): "Match the following select the correct option for the quantity of electricity, in Cmol^{-1} required to deposit various metals at cathode..." [Ans: N/A]
- [2024 Shift] Q#1 (MCQ): "How many coulombs are required to oxidise 0.1 mole of H_2O to oxygen?..." [Ans: N/A]

3.4 Electrode Potential & Delta G Relation (Moderate Yield — 4 PYQs)

Core Principle: Relationship between standard reduction potentials and Gibbs free energy: $\Delta G^\circ = -nFE^\circ$. When combining half-reactions, always add extensive ΔG values, never add intensive E° potentials directly!

■■■ **Caution / JEE Pitfall:** Directly adding E° potentials of two half-cells to obtain E° of a composite couple (e.g. $\text{Fe}^{3+} \rightarrow \text{Fe}$ from $\text{Fe}^{3+} \rightarrow \text{Fe}^{2+}$ and $\text{Fe}^{2+} \rightarrow \text{Fe}$). You MUST weight by electron transfer count: $n_3 E_3^\circ = n_1 E_1^\circ + n_2 E_2^\circ$!

■ **Speed Trick:** Always write $\Delta G = -nFE$ before combining half-reactions. A more positive standard reduction potential E° denotes a stronger oxidizing agent (more readily reduced).

Representative Source PYQs:

- [2022 Shift] Q#1 (MCQ): "For spontaneity of a cell, which is correct?..." [Ans: N/A]
- [2020 Shift] Q#3 (MCQ): "Given $E^\circ(\text{Fe}^{3+}/\text{Fe}^{2+}) = +0.76\text{V}$ and $E^\circ(\text{I}_2/\text{I}^-) = +0.55\text{V}$. The equilibrium constant for the reaction taking place in galvanic..." [Ans: N/A]
- [2019 Shift] Q#3 (MCQ): "Give $E^\circ(\text{Mn}^{7+}/\text{Mn}^{2+}) = 1.5\text{V}$ and $E^\circ(\text{Mn}^{4+}/\text{Mn}^{2+}) = 1.2\text{V}$, then $E^\circ(\text{Mn}^{7+}/\text{Mn}^{4+})$ is..." [Ans: N/A]

3.5 Kohlrausch's Law & Weak Electrolyte Dissociation (Standard Yield — 2 PYQs)

Core Principle: Independent migration of ions at infinite dilution: $\Lambda_m^\circ = \nu_+ \lambda_+^\circ + \nu_- \lambda_-^\circ$. Degree of dissociation $\alpha = (\Lambda_m / \Lambda_m^\circ)$, and Ostwald's dilution law $K_a = (\alpha^2 / (1 - \alpha))$.

■■■ **Caution / JEE Pitfall:** 1. Forgetting to multiply ionic conductances by stoichiometric coefficients (e.g., for $\text{Al}_2(\text{SO}_4)_3$ implies $2\lambda^\circ(\text{Al}^{3+}) + 3\lambda^\circ(\text{SO}_4^{2-})$). 2. For weak electrolytes, if $\alpha \ll 1$, do not approximate $1 - \alpha \approx 1$.

■ **Speed Trick:** Λ_m° for weak electrolytes like CH_3COOH can be obtained using strong electrolytes: $\Lambda_m^\circ(\text{CH}_3\text{COOH}) = \Lambda_m^\circ(\text{CH}_3\text{COONa}) + \Lambda_m^\circ(\text{HCl}) - \Lambda_m^\circ(\text{NaCl})$.

Representative Source PYQs:

- [2026 Shift] Q#3 (MCQ): " $\Lambda_m^\circ(\text{m}(\text{NH}_4\text{OH}))$ is equal to _____..." [Ans: N/A]
- [2019 Shift] Q#4 (MCQ): "Addition of excess of AgNO_3 to an aqueous solution of 1 mole of $\text{PdCl}_2 \cdot 4\text{NH}_3$ gives 2 moles of AgCl . The conductivity of ..." [Ans: N/A]

3.6 Commercial Batteries, Fuel Cells & Corrosion (Standard Yield — 1 PYQs)

Core Principle: Primary cells (dry cell, mercury cell), secondary lead-acid batteries (charge/discharge cycles), $\text{H}_2\text{-O}_2$ fuel cells (thermodynamic efficiency $\eta = (\Delta G / \Delta H)$), and electrochemical corrosion prevention.

■■■ **Caution / JEE Pitfall:** In a lead storage battery, the density of H_2SO_4 decreases during discharge because sulfuric acid is consumed and water is produced.

■ **Speed Trick:** In galvanic cells: Anode is negative, Cathode is positive (An Ox, Red Cat). In electrolytic cells: Anode is positive, Cathode is negative.

Representative Source PYQs:

- [2022 Shift] Q#3 (MCQ): "In fuel cells _____ are used as catalysts...." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

The conductivity of centimolar solution of KCl at 298 K is $0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and the resistance of the cell containing the solution at 298 K is 60Ω . The value of cell constant (G^*) is _____.

- (A) 3.28 cm^{-1}
- (B) 1.26 cm^{-1}
- (C) 3.34 cm^{-1} [Correct]
- (D) 1.34 cm^{-1}

Correct Answer: C

Solution Analysis: Conductivity is related to conductance and cell constant by $\kappa = G \times G^*$ where $G = (1/R)$ So, $\kappa = (1/R) \times G^*$. This gives $G^* = \kappa \times R$ Given, $\kappa = 0.021 \text{ Ohm}^{-1} \text{ cm}^{-1}$ and $R = 60 \Omega$ So, $G^* = 0.021 \times 60$ $G^* = 1.26 \text{ cm}^{-1}$ So, the correct option is 1.26 cm^{-1} Option B.

Benchmark Problem 2: KCET 2026 (Q#2)

Given below are the half-cell reactions: $\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$ ($E^\circ = -1.18 \text{ V}$) $2(\text{Mn}^{3+} + \text{e}^- \rightarrow \text{Mn}^{2+})$ ($E^\circ = +1.51 \text{ V}$) The E_{cell}° for $3\text{Mn}^{2+} \rightarrow \text{Mn} + 2\text{Mn}^{3+}$ will be _____.

- (A) -2.69 V , the reaction will not occur (Non-Spontaneous)
- (B) -2.69 V , the reaction will occur (Spontaneous)
- (C) -0.33 V , the reaction will not occur (Non-Spontaneous)
- (D) -0.33 V , the reaction will occur (Spontaneous)

Correct Answer: None

Solution Analysis: For the reaction $3\text{Mn}^{2+} \rightarrow \text{Mn} + 2\text{Mn}^{3+}$ one part of Mn^{2+} is getting reduced to Mn : $\text{Mn}^{2+} + 2\text{e}^- \rightarrow \text{Mn}$ $E^\circ = -1.18 \text{ V}$ So this acts as the cathode reaction. The other part of Mn^{2+} is getting oxidised to Mn^{3+} : $2\text{Mn}^{2+} \rightarrow 2\text{Mn}^{3+} + 2\text{e}^-$ This is the reverse of $2(\text{Mn}^{3+} + \text{e}^- \rightarrow \text{Mn}^{2+})$ $E^\circ = +1.51 \text{ V}$ So the oxidation potential will be $E_{\text{ox}}^\circ = -1.51 \text{ V}$ Now, $E_{\text{cell}}^\circ = E_{\text{cathode}}^\circ + E_{\text{anode}}^\circ$

Benchmark Problem 3: KCET 2026 (Q#3)

$\Lambda_{\text{m}}(\text{NH}_4\text{OH})^\circ$ is equal to _____

- (A) $\Lambda_{\text{m}}(\text{NH}_4\text{OH})^\circ + \Lambda_{\text{m}}(\text{NH}_4\text{Cl})^\circ - \Lambda_{\text{m}}(\text{HCl})^\circ$
- (B) $\Lambda_{\text{m}}(\text{NH}_4\text{Cl})^\circ + \Lambda_{\text{m}}(\text{NaOH})^\circ - \Lambda_{\text{m}}(\text{NaCl})^\circ$
- (C) $\Lambda_{\text{m}}(\text{NH}_4\text{Cl})^\circ + \Lambda_{\text{m}}(\text{NaCl})^\circ - \Lambda_{\text{m}}(\text{NaOH})^\circ$
- (D) $\Lambda_{\text{m}}(\text{NaOH})^\circ + \Lambda_{\text{m}}(\text{NaCl})^\circ - \Lambda_{\text{m}}(\text{NH}_4\text{Cl})^\circ$

Correct Answer: None

Solution Analysis: For NH_4OH , $\Lambda_{\text{m}}(\text{NH}_4\text{OH})^\circ = \lambda_{\text{NH}_4^+}^\circ + \lambda_{\text{OH}^-}^\circ$ Using Kohlrausch's law, we combine strong electrolytes whose ions can give NH_4^+ and OH^- . Now, $\Lambda_{\text{m}}(\text{NH}_4\text{Cl})^\circ = \lambda_{\text{NH}_4^+}^\circ + \lambda_{\text{Cl}^-}^\circ$ $\Lambda_{\text{m}}(\text{NaOH})^\circ = \lambda_{\text{Na}^+}^\circ + \lambda_{\text{OH}^-}^\circ$ $\Lambda_{\text{m}}(\text{NaCl})^\circ = \lambda_{\text{Na}^+}^\circ + \lambda_{\text{Cl}^-}^\circ$ So, $\Lambda_{\text{m}}(\text{NH}_4\text{OH})^\circ = \Lambda_{\text{m}}(\text{NH}_4\text{Cl})^\circ + \Lambda_{\text{m}}(\text{NaOH})^\circ - \Lambda_{\text{m}}(\text{NaCl})^\circ$

Benchmark Problem 4: KCET 2026 (Q#4)

During the electrolysis of acidified water, 16 g of O_2 gas is formed at anode. The volume of H_2 gas liberated at cathode under STP conditions is

- (A) 22.4 L **[Correct]**
- (B) 11.2 L
- (C) 2.24 L
- (D) 1.12 L

Correct Answer: A

Solution Analysis: During electrolysis of acidified water, hydrogen gas is formed at the cathode and oxygen gas is formed at the anode. The overall reaction is $2H_2O(l) \rightarrow 2H_2(g) + O_2(g)$. From the equation, 1 mole of O_2 gives 2 moles of H_2 . Given mass of oxygen gas: 16g. Molar mass of O_2 is $32g\ mol^{-1}$. So, moles of O_2 formed: $(16 / 32) = 0.5\ mol$. Since hydrogen is formed in double the number of moles, moles of $H_2 = 2 \times 0.5 = 1\ mol$. At STP, 1 mole of any gas occupies 22.4L. So, volume of H_2 gas liberated is

Benchmark Problem 5: KCET 2020 (Q#3)

Given $E_{(Fe^{3+}/Fe^{2+})^o} = +0.76V$ and $E_{(I_2/I^-)^o} = +0.55V$. The equilibrium constant for the reaction taking place in galvanic cell consisting of above two electrodes is $[(2303RT / F) = 0.06]$

- (A) 1×10^7
- (B) 1×10^9
- (C) 3×10^8
- (D) 5×10^{12}

Correct Answer: None

Solution Analysis: $E_{(Fe^{3+}/Fe^{2+})^o} = 0.76V$ and $E_{(I_2/I^-)^o} = 0.55V$. $K = ?$ Also $(2.303RT / F) = 0.06$. Here, I_2/I^- will act as anode and Fe^{3+}/Fe^{2+} will act as cathode. $2Fe^{3+} + 2I^- \rightarrow 2Fe^{2+} + I_2$. Here, $n = 2$. $E_{(cell)^o} = E_{(cathode)} - E_{(anode)} = 0.76 - 0.55 = +0.21$. $\Delta G^o = -nFE_{(cell)^o} = -2(96500) \times 0.21 = -40530J$. Also, $\Delta G = -2.303RT \log K_{(c)}$.

$$\log K_{(c)} = \frac{-\Delta G}{2.303RT} = \frac{-(-40530)}{(2 \times 2.303 \times 298)} = \frac{40530}{1365.72} = 29.69$$